

# DARSINI LAKSHMIAH

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Data Scientist with 4 years of experience building production machine learning and GenAI solutions, data pipelines, relational data models, and model monitoring systems across retail, marketing, and financial services. Experienced in Python, SQL, PySpark, MLflow, cloud data platforms, model-result evaluation, and API-oriented deployment workflows, with a track record of translating complex analyses into reliable decisions.

## WORK EXPERIENCE

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### AARP, Washington, DC, USA

Data Scientist

September 2025 - March 2026

- Architected a RAG marketing intelligence assistant for **1K+** campaign briefs, audience research, performance reports, and experiment readouts using Llama, embeddings, hybrid BM25/vector retrieval, metadata filtering, and reranking; organized source metadata and model outputs to deliver cited answers that accelerated campaign planning.
- Established a **200+** query evaluation framework for machine learning model results, tracking Recall@5, MRR, groundedness, and citation accuracy in MLflow; improved retrieval recall by **~15%** through chunking, hybrid search, and reranking.
- Trained and operationalized a LightGBM propensity model across **20M+** email, web, and campaign interactions, engineering recency, frequency, channel, affinity, and response features in PySpark; integrated eligibility, contact-history, and suppression rules into repeatable scoring workflows for **20+** campaigns, reducing manual preparation by **~30%**.
- Containerized a RAG-based analytics service using Docker and exposed retrieval and LLM workflows through REST APIs, packaging application dependencies and configuration for reproducible deployment; implemented authentication, logging, health checks, and response monitoring to support reliable internal usage.

### AMAZON, Austin, Texas, USA

Business Intelligence Engineer

May 2025 - August 2025

- Designed a unified customer-view data layer by cleaning and consolidating **87M+** Amazon Fresh and Whole Foods transactions from **530+** stores using SQL, Python, AWS Glue, and Spark; established structured data foundations for segmentation, behavioral analysis, and product-adoption measurement.
- Evaluated post-launch Amazon Internal Product behavior through longitudinal cohort and difference-in-differences analysis, measuring order frequency, spend, retention, and cross-banner migration; automated recurring measurement workflows in SQL and AWS and surfaced statistically significant opportunities to product leadership.
- Operationalized reusable customer-segmentation and KPI data pipelines in Redshift and AWS, standardizing **15+** definitions across orders, basket size, retention, Prime penetration, and channel adoption; replaced one-off analyses with a recurring measurement framework for AFS/WFM product launches.

### NATWEST GROUP, Chennai, India

Senior Data Scientist

Aug 2021 - June 2024

- Engineered fraud-detection machine learning models using XGBoost, TF-IDF, and NLP on **1.5M+** daily card, ACH, and digital banking transactions, reducing false positives by **12%** through feature engineering and hyperparameter tuning.
- Built standardized model inventory and monitoring workflows for **30+** production fraud models using Python, SQL, and MLflow, tracking model versions, metadata, evaluation results, precision/recall, score drift, and feature stability; integrated monitoring outputs into structured tables to support reproducible model governance and reduced model-drift incidents by **20%** through early detection.
- Optimized SQL Server queries and Snowflake data models on Azure, improving query performance by **35%**; orchestrated staging and integration data pipelines with Python, PySpark, and Azure Data Factory to process **1.5M+** daily transactions.

## EDUCATION

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**George Washington University, Washington, DC** - Master of Science in Data Science, GPA - 3.97/4.0 **May 2026**

**Madras Christian College, Chennai, India** - Bachelor of Science in Mathematics, CGPA - 8.50/10 **May 2021**

## PROJECTS

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**AI Interview Coach** | [GitHub](#) | [Website](#) | AutoGen · GPT-4o · FastAPI · Whisper · WebSocket · face-api.js

- Built a real-time multi-agent interview simulation where AI interviewer, evaluator, and candidate agents conduct company-specific mock interviews over WebSocket with live Whisper transcription and webcam-based performance scoring.

**NutriNav Food Access Intelligence App** | [GitHub](#) | [DevPost](#) | XGBoost · Groq Whisper · FastAPI · Supabase

- Top 3 of 35 teams, George Hacks × UN Reboot the Earth - judges highlighted immediate deployability and end-to-end integration. Built a multilingual voice-first app connecting 75,000+ DC SNAP residents to real-time corner store inventory with nutrition scoring via USDA data.

## CERTIFICATIONS

- Databricks Certified Generative AI Engineer
- Microsoft Certified: Fabric Analytics Engineer
- AWS Certified AI Practitioner
- Microsoft Certified: Machine Learning Operations Engineer

## TECHNICAL SKILLS

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- **Programming Languages:** Python, R, SQL, SAS, PySpark
  - **ML & AI:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, MLflow, SageMaker, LangChain, Machine Learning Models
  - **Cloud & Data Engineering:** AWS (Glue, Redshift, Athena, S3, Lambda, Bedrock), Azure Data Factory, ADLS Gen2, Snowflake, Airflow, Databricks, Data Pipelines
  - **Databases & Vector Stores:** MySQL, ChromaDB, SQL Server, Relational Database Design
  - **Analytics & Modeling:** A/B Testing, Causal Inference, NLP, KPI Reporting, Data Modeling, MLOps, Time Series Forecasting
  - **Visualization & Tools:** Tableau, Power BI, Looker, Pandas, NumPy, Git/GitHub, Docker, Flask
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